Problem Statement 3

HITACHI

AI Assisted OT Security Monitoring

Short Description

Industrial Control Systems (ICS) that operate critical infrastructure—such as manufacturing lines, energy systems, and water treatment plants—rely on Operational Technology (OT) networks using legacy protocols like Modbus, DNP3, and EtherNet/IP. These systems were designed for reliability and real-time control, not cybersecurity. As IT–OT convergence increases connectivity for analytics and remote operations, the attack surface of industrial environments expands significantly.

The challenge is to develop a lightweight AI-assisted OT security monitoring prototype that can analyze industrial network traffic or process telemetry and identify suspicious behavior. Using open-source ICS tools and datasets (e.g., OpenPLC, Conpot, Zeek ICS analyzers), participants should build a system that detects anomalies, interprets potential threats, and presents clear insights to operators through a simple dashboard.

Resource Availability

Open data sets
•(BATADAL)
https://www.batadal.net/data.html
•(TON_IoT datasets)
https://research.unsw.edu.au/projects/toniot-datasets

Reference:
[1] Zahoor, A., Abbasi, W., Babar, M.Z. *et al.* Robust IoT security using isolation forest and one class SVM algorithms. *Sci Rep* **15**, 36586 (2025). https://doi.org/10.1038/s41598-025-20445-4

Objectives

The solution should demonstrate how **AI can enhance situational awareness and accelerate cyber incident detection in industrial environments** without disrupting existing OT infrastructure.

Deliverables / Outcome

- Functional prototype (mandatory) with report including architecture description, algorithms used.
- Results benchmarking obtained results with those reported in earlier research.

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